ACTVET
Abu Dhabi Centre for Technical and Vocational Education and Training
مركز أبوظبي للتعليم والتدريب التقني والمهني

مدارس التكنولوجيا التطبيقية
Applied Technology Schools

AI SafeGuard



SMART SAFETY,

REAL PROTECTION.

Team Members & Roles



Sara Saoud Aljenibi

20230016864

Design and UI Developer

Designs the user interface and creates a clear, accessible dashboard experience.



Aisha Abdulla Alshehyari

572356

System Development and Research

Developed the system, integrated AI, and managed research, data collection, and analysis.



Salamah Ahmed Alyammahi

814091

Quality Testing and Support

Tests the system, documents problems and helps evaluate its accuracy and performance.



Idea Generation

AI SafeGuard

An AI-based workplace safety system that uses computer vision and facial recognition to identify workers and check whether they are wearing the required personal protective equipment (PPE).

Camel Watch AI

An AI-based road-safety system that detects camels and wildlife near desert highways and sends early warnings to drivers to help prevent accidents.

Chosen Idea : AI SafeGuard

We chose AI SafeGuard, an AI-powered workplace safety monitoring system, for the following reasons:

01 the system improves safety by checking whether workers are wearing the required protective equipment before entering workshops, laboratories, or industrial areas.

02 AI SafeGuard provides real-time monitoring and immediate alerts. This allows supervisors to respond quickly to safety violations and prevent possible accidents.

03 the project combines object detection, facial recognition, backend development, and dashboard design, allowing us to apply different computing skills in one practical system.



Problem Statement

Workers in workshops, laboratories and industrial environments must wear personal protective equipment to reduce the risk of injury. However, some workers may forget or ignore these requirements.

Traditional monitoring depends heavily on supervisors, who may miss violations because of distractions, large work areas or demanding workloads. Manual incident recording can also be slow and inconsistent.

Supervisors therefore need a reliable system that can check safety compliance in real time, identify workers and maintain accurate incident records. AI SafeGuard addresses this need through automated PPE detection, facial recognition, immediate alerts and incident logging.



Project Overview

AI SafeGuard is an AI-powered monitoring system designed for workshops, laboratories and industrial workplaces.

A webcam captures an image of a worker at an entry point. The system uses a custom YOLO11s model to detect visible safety equipment and DeepFace facial recognition to identify the worker. It then compares the detected equipment with the requirements configured for that entry point.

If the worker meets all requirements, the system displays a compliant result. If equipment is missing, it produces an audible alert and records the worker’s identity, entry point, missing equipment and scan image.

Supervisors can use the dashboard to review incidents, manage workers and entry points, and monitor compliance information.



Conducted Research

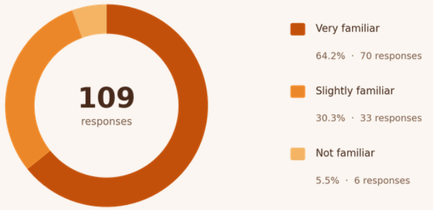
Background Research :
Research has explored using cameras and AI to detect missing safety gear and unsafe actions (Adhikesaven, 2022).

Existing Solution :
Intenseye uses cameras to monitor safety gear, send alerts, and track safety compliance. AI SafeGuard focuses on a webcam-based system for laboratory and workshop entrances.

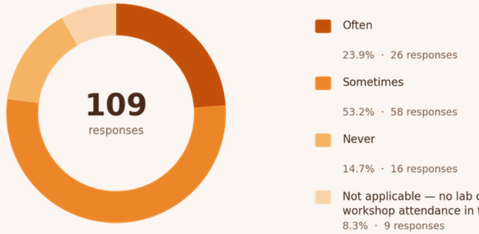
User Research :
Our survey was shared with students and received 109 responses:

- 77% had seen someone enter without required safety gear often or sometimes.
- 78% considered missing safety gear a serious problem.
- 95% thought automatic safety-gear checks would be useful.
- 79% supported a supervised school trial.

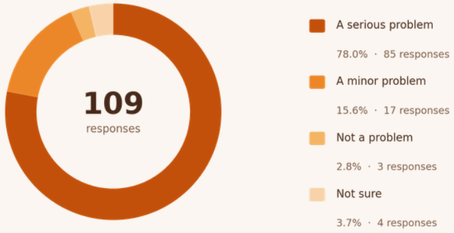
01 Safety-gear familiarity



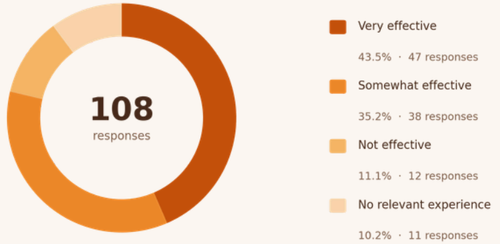
02 Observed missing safety gear



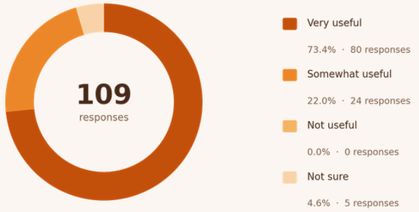
03 How serious is the problem?



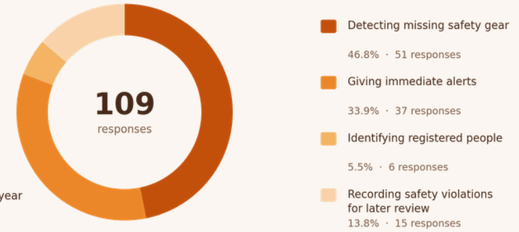
04 Teacher / supervisor effectiveness



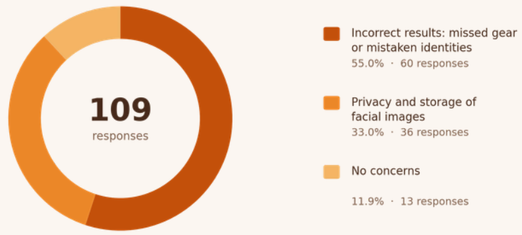
05 Usefulness of automatic checks



06 Most valuable feature



07 Main concern



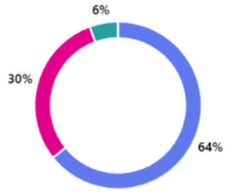
08 Support for a supervised trial



Data Analysis

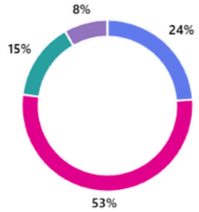
1. **How familiar are you with wearing safety gear, such as safety goggles, gloves, or lab coats, during school laboratory or workshop activities?** More details

Very familiar	70
Slightly familiar	33
Not familiar	6



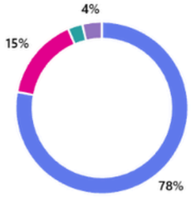
2. **In the past year, how often have you seen someone enter a laboratory or workshop without the required safety gear?** More details

Often	26
Sometimes	58
Never	16
Not applicable - I have not attended either during this period	9



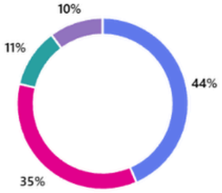
3. **In your opinion, how serious is the problem of people entering laboratories or workshops without the required safety gear?** More details

A serious problem	85
A minor problem	17
Not a problem	3
Not sure	4



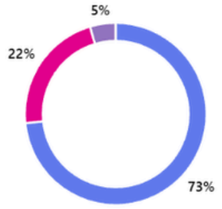
4. **Based on your experience, how effective are teachers or supervisors at noticing missing safety gear?** More details

Very effective	47
Somewhat effective	38
Not effective	12
No relevant experience	11



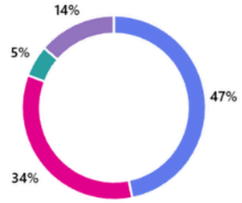
5. **How useful would an automatic safety-gear check at a laboratory or workshop entrance be?** More details

Very useful	80
Somewhat useful	24
Not useful	0
Not sure	5



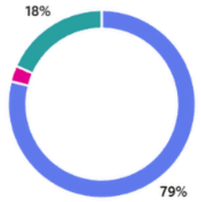
6. **Which proposed AI SafeGuard feature would be most valuable to you?** More details

Detecting missing safety gear	51
Giving immediate alerts	37
Identifying registered people	6
Recording safety violations for later review	15



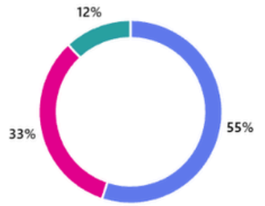
8. **Would you support a supervised trial of AI SafeGuard in a school laboratory or workshop?** More details

Yes	86
No	3
Maybe	20



7. **What would be your main concern about AI SafeGuard?** More details

Incorrect results, such as missed safety gear or mistaken identities	60
Privacy and storage of facial images	36
No concerns	13



What the Results Mean for AI SafeGuard :

Among 109 respondents, 95% considered automatic checks useful. Detecting missing gear (47%) and immediate alerts (34%) were the most preferred features, supporting our focus on these functions.

Incorrect results (55%) and facial-image privacy (33%) were the main concerns, highlighting accuracy testing and protection of facial data as development priorities.

Resources and Materials Used

01

AI and Image Processing

The active PPE model uses Ultralytics YOLO11s with PyTorch. OpenCV processes camera images, while YuNet detects faces and Facenet512 generates facial embeddings for worker identification. DeepFace supports the facial-recognition workflow.

02

Frontend and Backend Development

The frontend uses React, Vite, HTML, CSS and JavaScript. The FastAPI backend manages image uploads, AI processing, worker records and incident data, while Uvicorn runs the application locally.

03

Data and Training

Version 1 used approximately 68 curated Roboflow datasets and was trained in Kaggle with an NVIDIA Tesla T4 GPU. Version 2 used locally recorded camera videos with manually reviewed annotations.

04

Hardware

The required hardware consists of a development laptop and an external USB webcam.

Challenges and Solutions

Version 2 missed some equipment under our webcam conditions. We recorded local videos, manually corrected the annotations, and fine-tuned the model, increasing frame-level recall from 45% to 100% on our separate test video.

Training the YOLO model locally was very slow and caused the laptop to overheat. We moved the training process to Kaggle and used an NVIDIA Tesla T4 GPU to complete it reliably.

The public datasets used different class names and annotation formats. We cleaned and combined approximately 68 datasets, corrected annotation problems, and standardized them into six PPE classes.

RetinaFace took 9–16 seconds to detect a face, so we replaced it with YuNet. This reduced face-detection time to approximately 0.05–0.10 seconds while keeping Facenet512 for identity matching.

Risk Assessment

Incorrect PPE Detection

Poor lighting, hidden equipment, or unusual camera angles may affect detection accuracy, so the camera should provide a clear view and the system should be tested under different conditions.

Incorrect Worker Identification

Facial recognition may fail when a face is covered or unclear, so the system uses a strict matching threshold and allows supervisors to verify uncertain results.

Privacy and Data Security

Worker photographs and incident records contain sensitive information, so access should be restricted and all data should be stored through protected backend routes.

Overreliance on AI

AI can produce incorrect results, so AI SafeGuard should support safety supervision rather than replace it. A supervisor should review incidents and make final decisions.

Development Phases

1. **Research and Planning-** We identified the limitations of manual PPE monitoring and defined the project requirements.

2. **Interface Design-** We designed and developed the interface in Visual Studio Code using React and Vite.

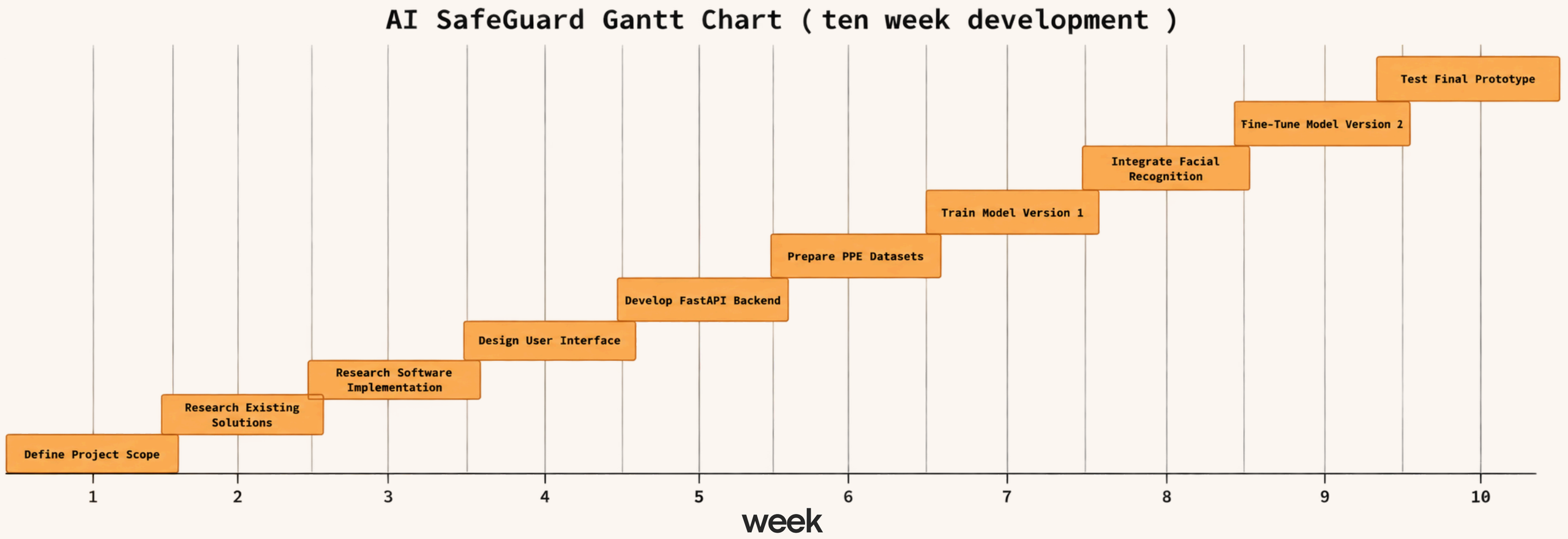
3. **Backend Development-** We built a FastAPI backend for image uploads, worker registration, scan processing, and incident storage.

4. **PPE Model Development-** We created several YOLO model versions and improved the dataset, class structure, and annotations after each test.

5. **Facial-Recognition Integration-** We connected DeepFace and Facenet512, then replaced RetinaFace with YuNet to improve speed.

6. **Testing and Improvement-** We tested complete scans, corrected application errors, fine-tuned Version 2, and compared it with Version 1.

Gantt Chart


AI SafeGuard Gantt Chart (ten week development)
Define Project Scope
Research Existing Solutions
Research Software Implementation
Design User Interface
Develop FastAPI Backend
Prepare PPE Datasets
Train Model Version 1
Integrate Facial Recognition
Fine-Tune Model Version 2
Test Final Prototype
1
2
3
4
5
6
7
8
9
10
week

AI Implementation

The AI SafeGuard project integrates artificial intelligence to improve workplace safety through automatic PPE detection and worker identification.

PPE Detection

The system captures an image using the webcam and sends it to the YOLO11s model. The model analyzes the image and detects six types of PPE: helmets, goggles, gloves, safety vests, ear protection, and lab coats.

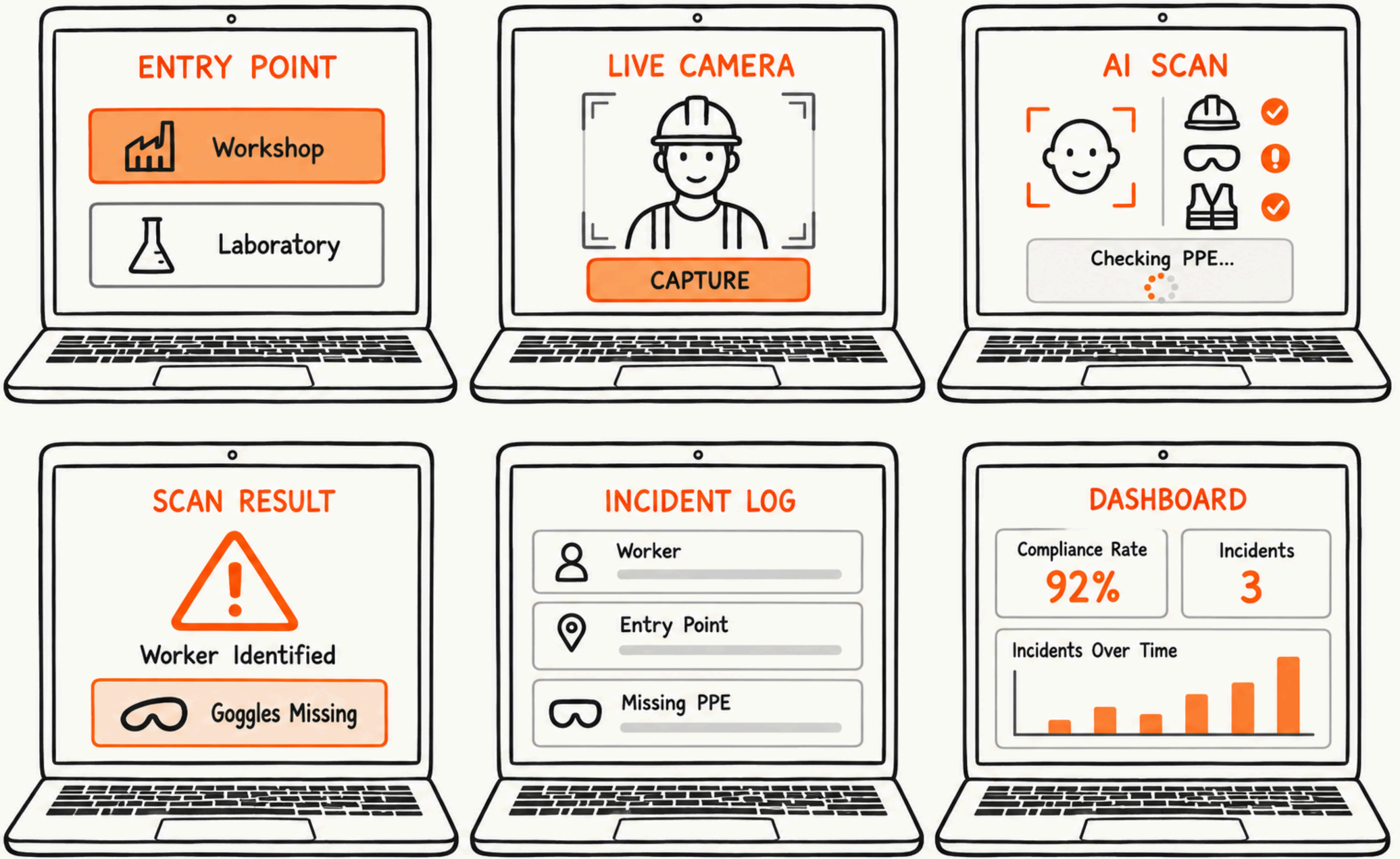
Facial Recognition

YuNet locates the worker's face in the captured image. Facenet512 then creates a facial embedding and compares it with the saved embeddings of registered workers to determine the worker's identity.

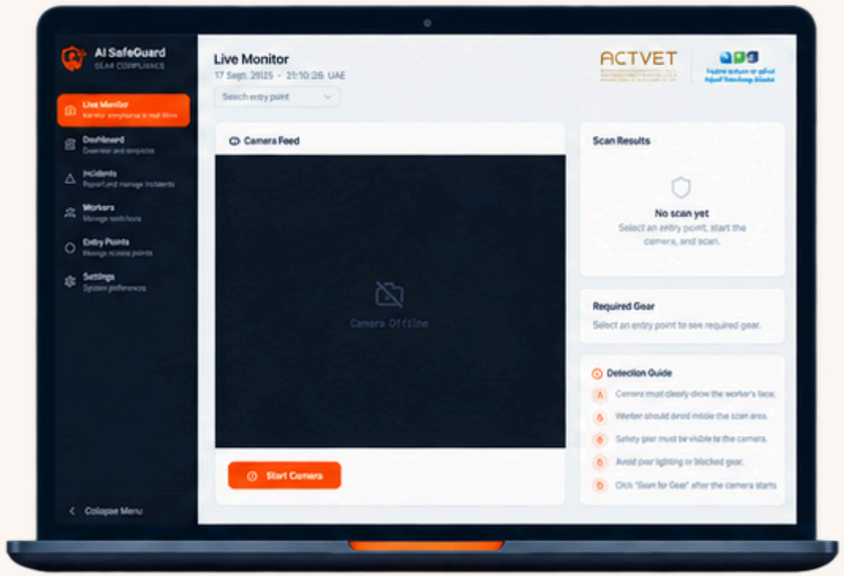
Compliance Decision

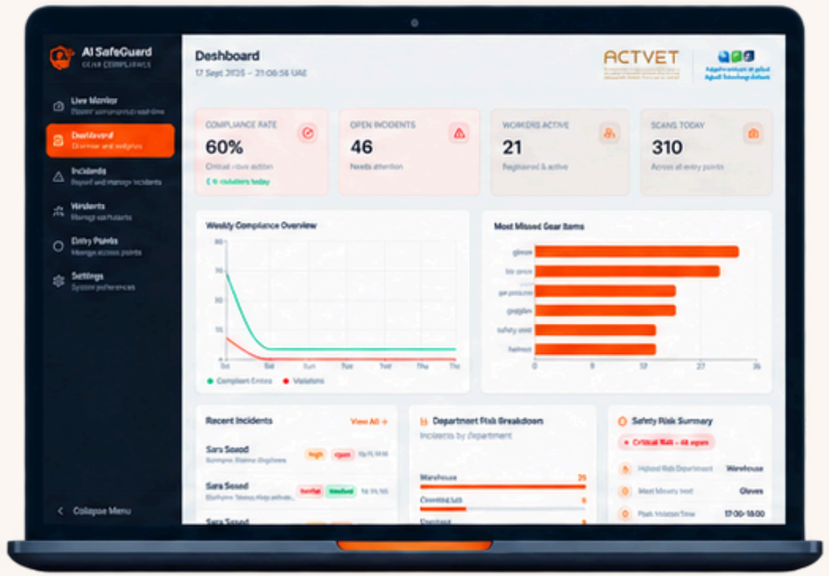
The system compares the detected PPE with the safety equipment required at the selected entry point. It then displays a compliant or violation result, identifies any missing equipment, and records the incident when necessary.

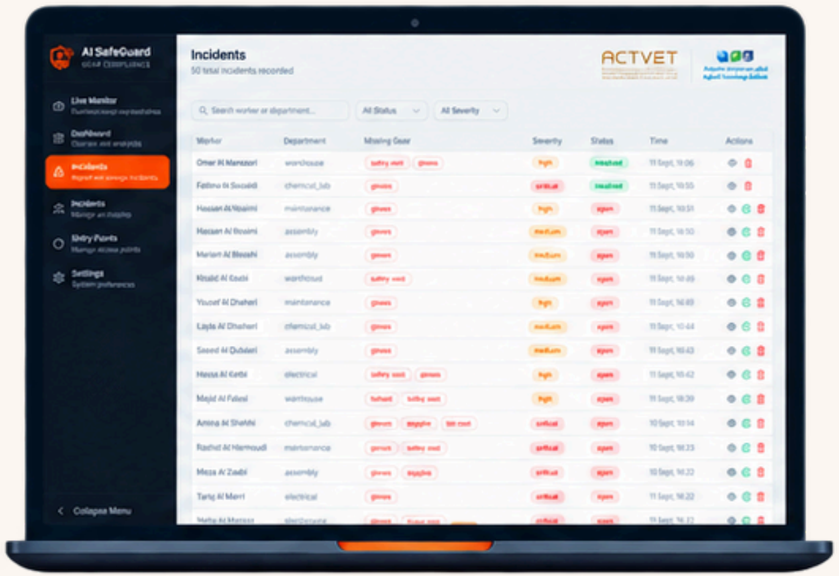
Initial Prototype

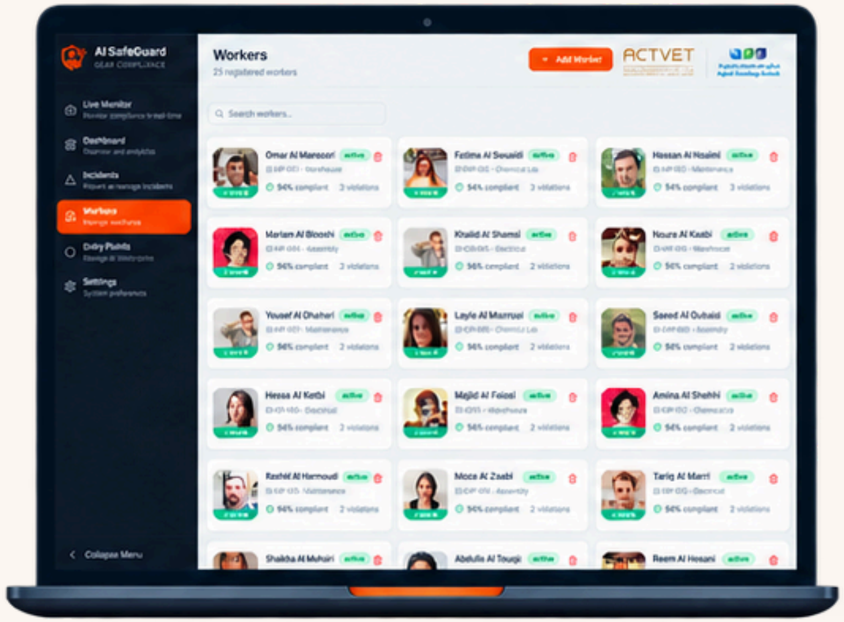


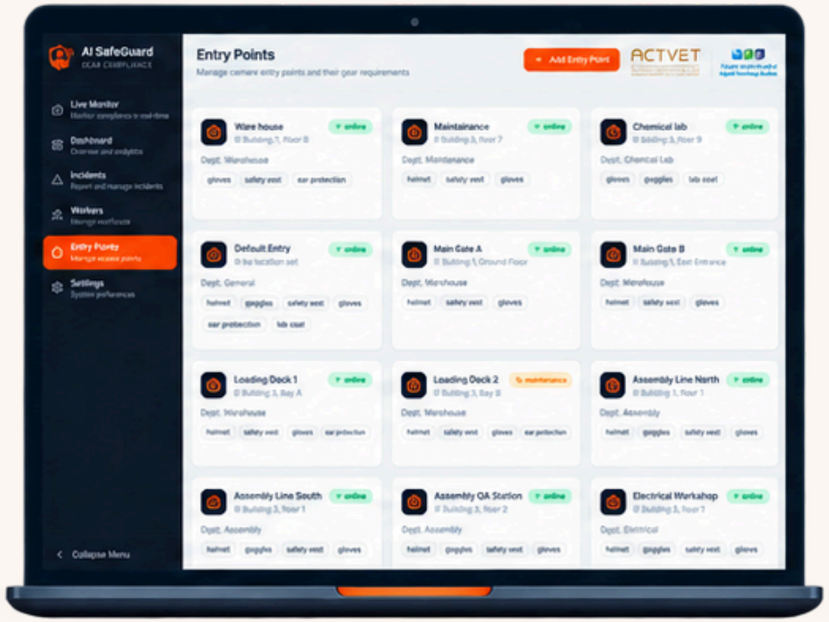
Final Prototype

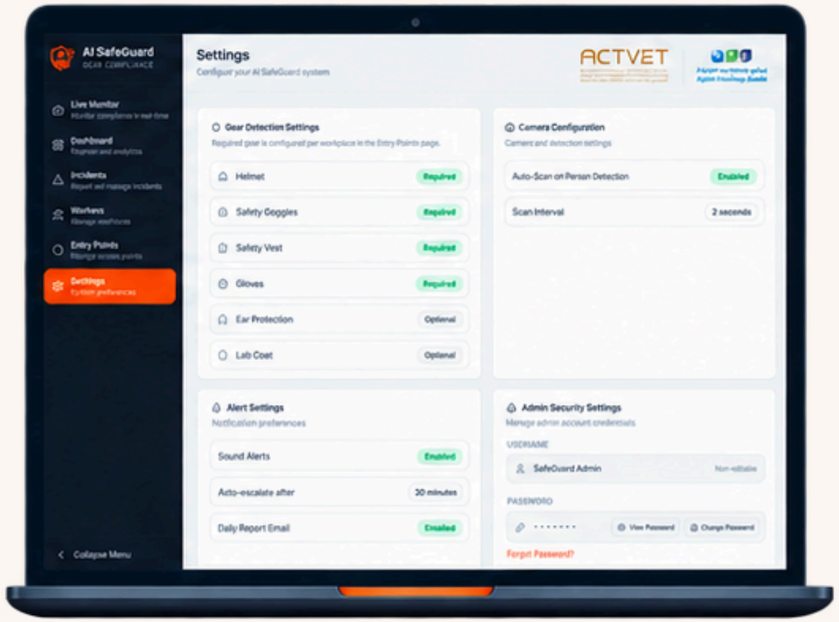












Future Plan

Deploy AI SafeGuard online so supervisors can access the system from different devices and locations.

Expand the training dataset to improve PPE detection accuracy in unfamiliar environments and difficult conditions.

Connect a secure cloud database to store worker information, incident records, and captured images safely.

Improve the system so it can detect and identify multiple workers within the same camera frame.

Test the system in real workplaces using different people, lighting conditions, camera angles, and PPE designs.

Add automatic email or mobile alerts to notify supervisors immediately when a safety violation occurs.

Conclusion

AI SafeGuard addresses the problem of missed PPE violations by automatically detecting safety equipment, identifying registered workers, and recording incidents in real time. Our final model achieved 95.9% recall and detected all six PPE classes in our separate test video.

Throughout this project, we learned how to prepare datasets, train AI models, develop a complete application, and improve it through testing.

The project shows how artificial intelligence can support supervisors, improve safety compliance, and help create safer workplaces.



Thank you very much for your attention!

